Unusual Billing Patterns in Medicaid Behavioral Health Services

Temporal Spikes, Outliers, and Specialty Mismatches in Behavioral Codes (Medicaid provider spending 2018 – 2024)

By Kelvin Nlebemchukwu

Executive Summary

The Department of Health and Human Services' DOGE recently released an unprecedented 11GB Medicaid dataset containing provider-level, time series claims information. Leveraging this dataset, this analysis identifies systemic risk patterns, high-cost service concentrations, and statistically irregular billing behaviors that mirror known fraud cases such as the large-scale autism-related scheme previously uncovered in Minnesota.

The investigation began by profiling the highest paid Medicaid providers. The top twenty providers collectively accounted for tens of billions in spending, with the single highest-paid provider receiving over \$7.1 billion. Service level analysis revealed extreme concentration in HCPCS code T1019 (personal care services), which alone exceeded \$16 billion in payments. Strong positive correlations between total paid, total claims, and unique beneficiaries suggest that high-volume providers drive a disproportionate share of Medicaid expenditures.

A focused review of nine high-risk behavioral and autism-related HCPCS codes (97151 - 97158, 0362T) uncovered more than 39 million claims totaling \$7.02 billion. Code 97153 emerged as the dominant cost driver, representing over \$5.1 billion in spending. Monthly trend analysis for 97153 revealed sharp, unsustainable spikes among several providers, classic indicators of potential "bust-out" fraud, rapid volume inflation, and volatility inconsistent with legitimate service delivery capacity.

Outlier detection at the per-beneficiary level further exposed providers billing 3× to 15× above median norms in both paid amounts and claim counts. These patterns strongly suggest inflated service intensity or billing misaligned with patient need.

A broader statistical mismatch analysis identified 291,753 irregular billing events where providers categorized as non-specialists who are billing specialty codes that are outside their defined scope or typical practice area, representing less than 1% of their typical service mix yet accounted for massive financial outlays. Community/Behavioral Health Agencies (251X) and Clinic/Center taxonomies (261X) were the most frequent sources of high-cost mismatches, particularly for T1019, T1015, and S5125.

Integrating statistical outliers, specialty mismatches, and temporal anomalies produced a ranked list of the top 300 highest-risk providers. The highest-risk provider accumulated over 37,000 risk points and more than \$4.5 billion in total paid claims. Geographic clustering revealed concentrated risk in California, Florida, and Ohio, with Mississippi exhibiting the highest average risk score per provider, more than double any other state.

Overall, the analysis demonstrates that Medicaid spending risk is heavily concentrated among a small subset of providers, specific behavioral health codes, and taxonomy-code combinations that deviate sharply from normative patterns. These findings validate DOGE's X formerly twitter assertion that the dataset can surface large-scale fraud signals and provide a foundation for targeted audits, enhanced oversight, and proactive fraud detection.

Methodology

This analysis aimed to identify and characterize high-risk Medicaid providers based on their billing patterns, combining large-scale claims data with provider demographic and specialty information. The methodology involved several distinct phases, leveraging Polars for efficient data processing and Matplotlib for visualization.

The primary dataset was approximately 11GB in size, detailed provider information was sourced from the NPPES data. This multi-gigabyte dataset was also loaded as a Polars Lazy Frame. Key columns, including NPI, Entity Type, Healthcare Provider Taxonomy Code, and practice location details were selected. The two datasets were combined using a left join operation on the respective NPI columns (Billing Provider NPI from Medicaid data and NPI from NPPES data). Enriching each Medicaid claim record with detailed provider attributes.

Three primary risk flags were defined and computed to identify irregular billing patterns which are statistical Outliers were total paid exceeding the 99th percentile of all total paid values in the dataset were flagged **(exhibit 1)**. This threshold was calculated using a preliminary collect (), call on the Medicaid data. Secondly, instances where specific high-risk HCPCS codes (e.g., autism treatment codes 97151 - 97158) were billed by providers whose taxonomy code did not align with a specialized behavior analyst taxonomy (103K00000X) were flagged. This identifies potential out of scope billing. Thirdly, temporal Spikes where monthly claim volumes "Total Claims" exceeding three times a provider's historical average monthly claims (calculated per provider across all their records) were flagged. This captures sudden, unusual surges in billing activity. The flagged records (those with risk score > 0) were aggregated by unique provider identifiers, along with their associated state, city, and taxonomy code. For each provider, the total paid, total claim, and counts for each individual flag type were summed. A total risk points metric was calculated by summing the risk score for all their flagged records.

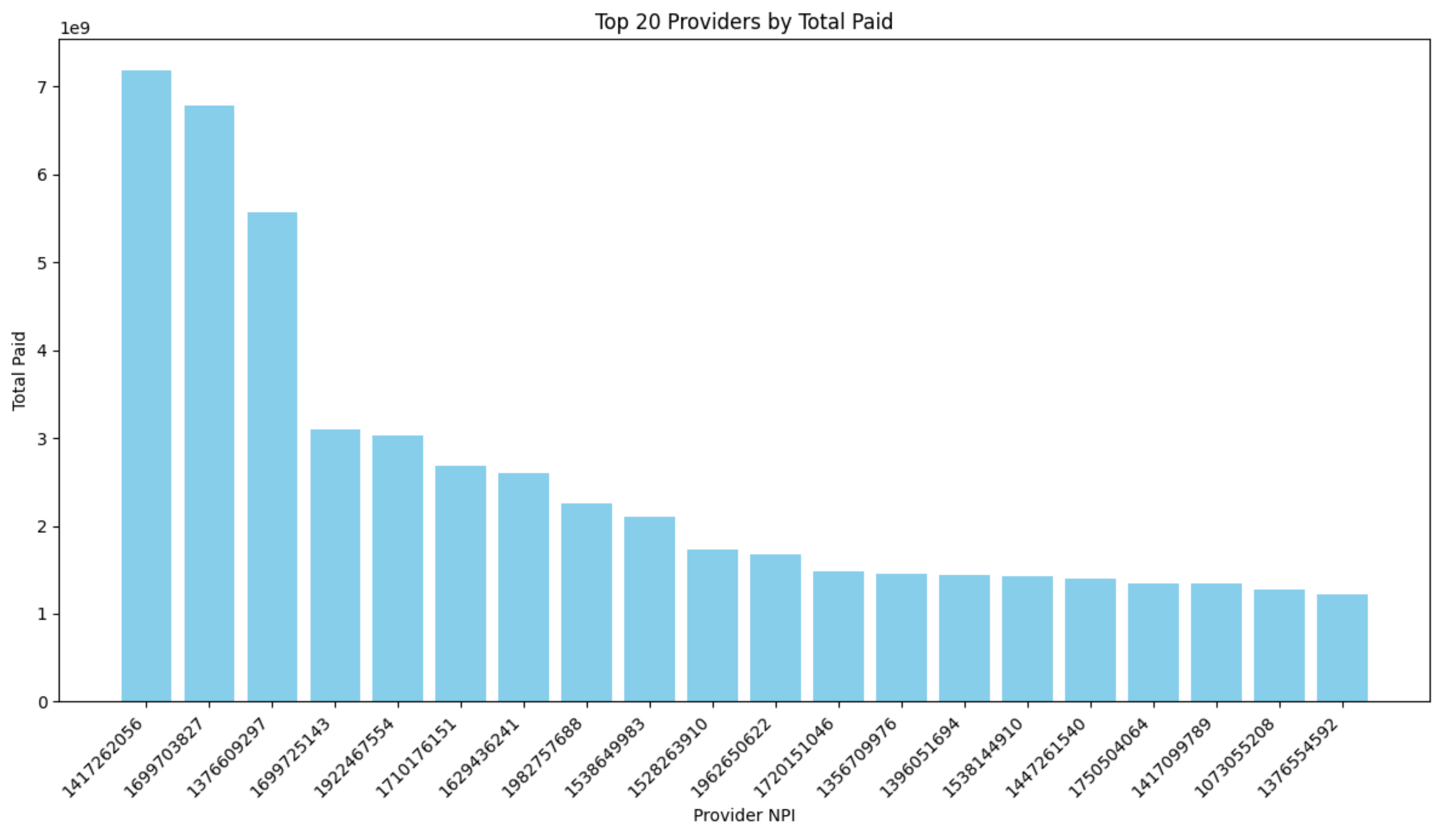
Providers were then ranked by total risk points in descending order, and the top 300 highest-risk providers were extracted to form the top 300 high risk providers **(exhibit 1)**

To provide comprehensive context for auding, detailed NPI attributes, including street addresses, were extracted from the NPPES dataset for the top 300 high-risk providers, forming the "Top 300 providers details" **(Exhibit 3).**

Key Findings

1. High-Risk Service Identification

Figure 1 Top 20 Providers by Total Paid

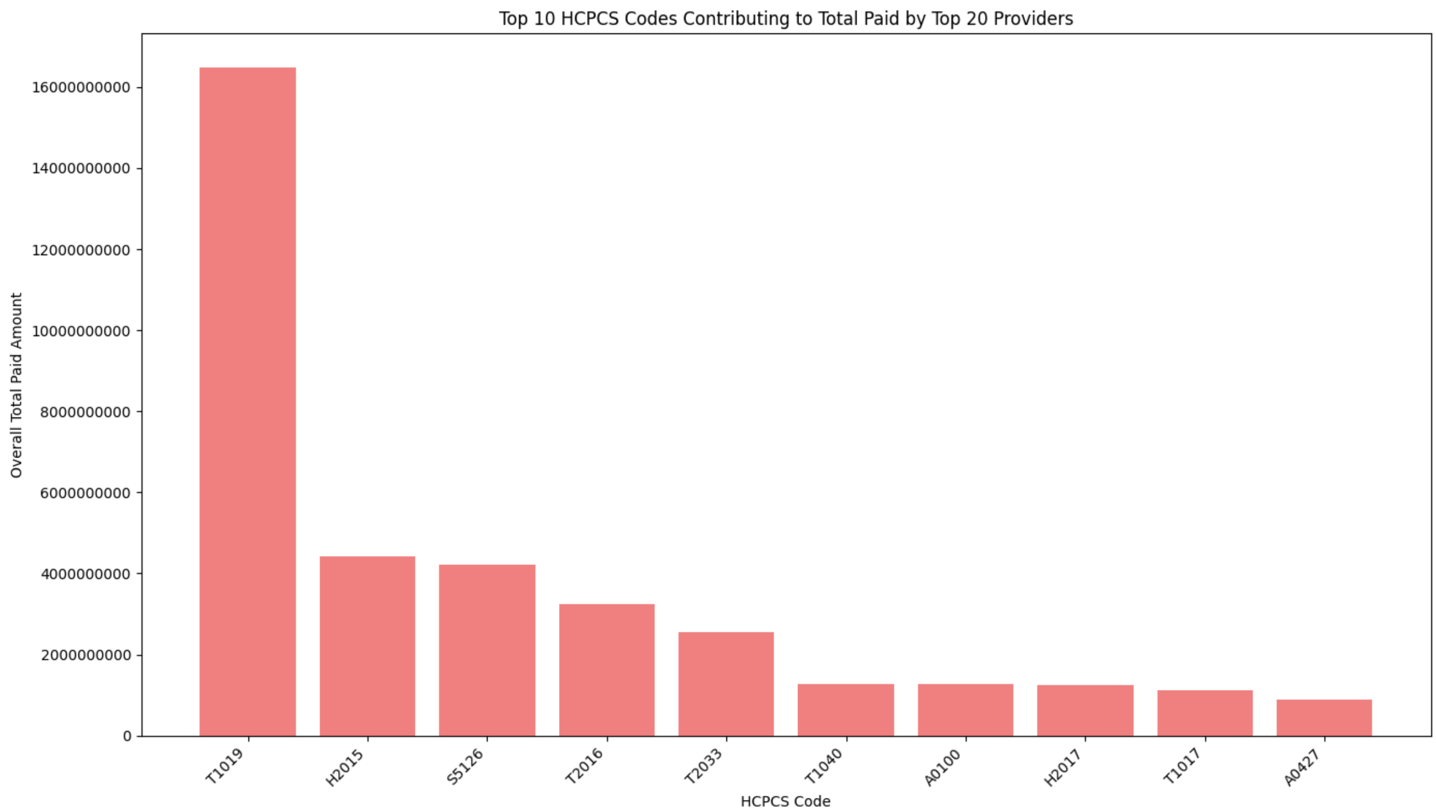


SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date Feb 15, 2026).

Figure 1 above shows the top 20 billing providers by total paid amount. The highest paid provider with NPI 1417262056 accounted for approximately $7.1778 billion in total paid claims and provider with 1376554592 ranked 20^{ths} with total payment received amounting to $1.2250 billion.

Further analysis to establish the services associated with the top paid providers to understand factors contributing to their high claim volumes and total paid amounts was carried out.

Figure 2 Top 10 HCPC codes contributing to total paid by top 20 providers

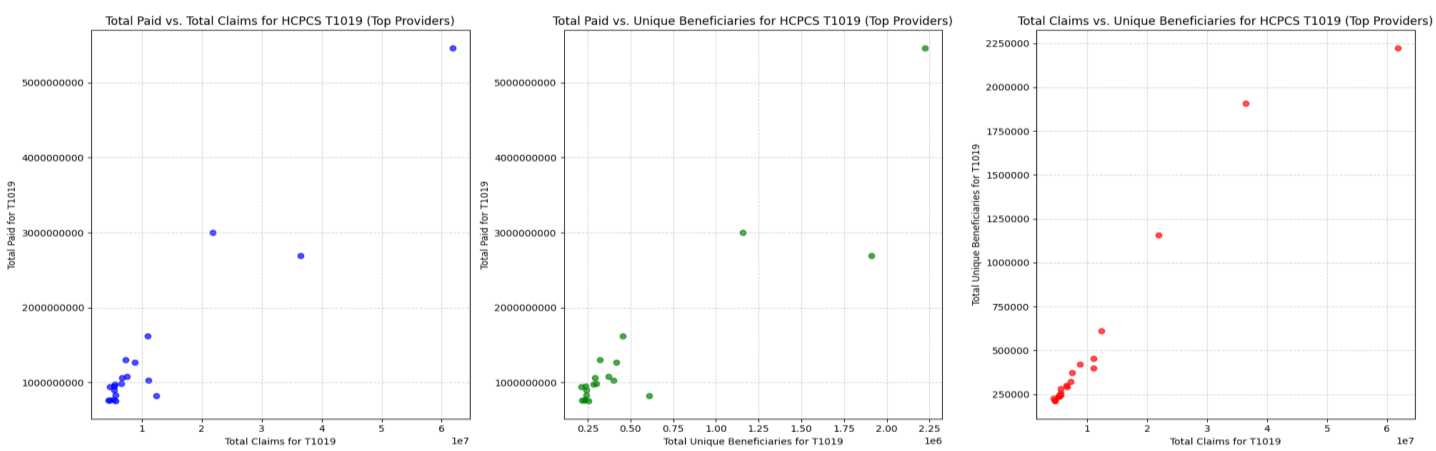


SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date Feb 15, 2026).

Figure 2 above shows the the Top 10 HCPCS code contributing to total paid by top 20 providers. Service T1019 ranked 1st with overall total paid amounting to over $16 billion while service A0427 ranked 10th with total paid amounting to less than $2 billion.

I carried out further analyses to explore the relationship between Total paid, Total claims, and Total Unique Beneficiary for HCPCS code T1019 to understand if higher payments correlate with more claims or more unique beneficiaries.

Figure 3 The analysis of the top 20 providers



SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date Feb 15, 2026).

The analysis of the top 20 providers for T1019 in figure 3 above reveals a highly concentrated distribution of services. The top provider alone accounts for approximately $5.4632 billion in total paid amounts for this code.

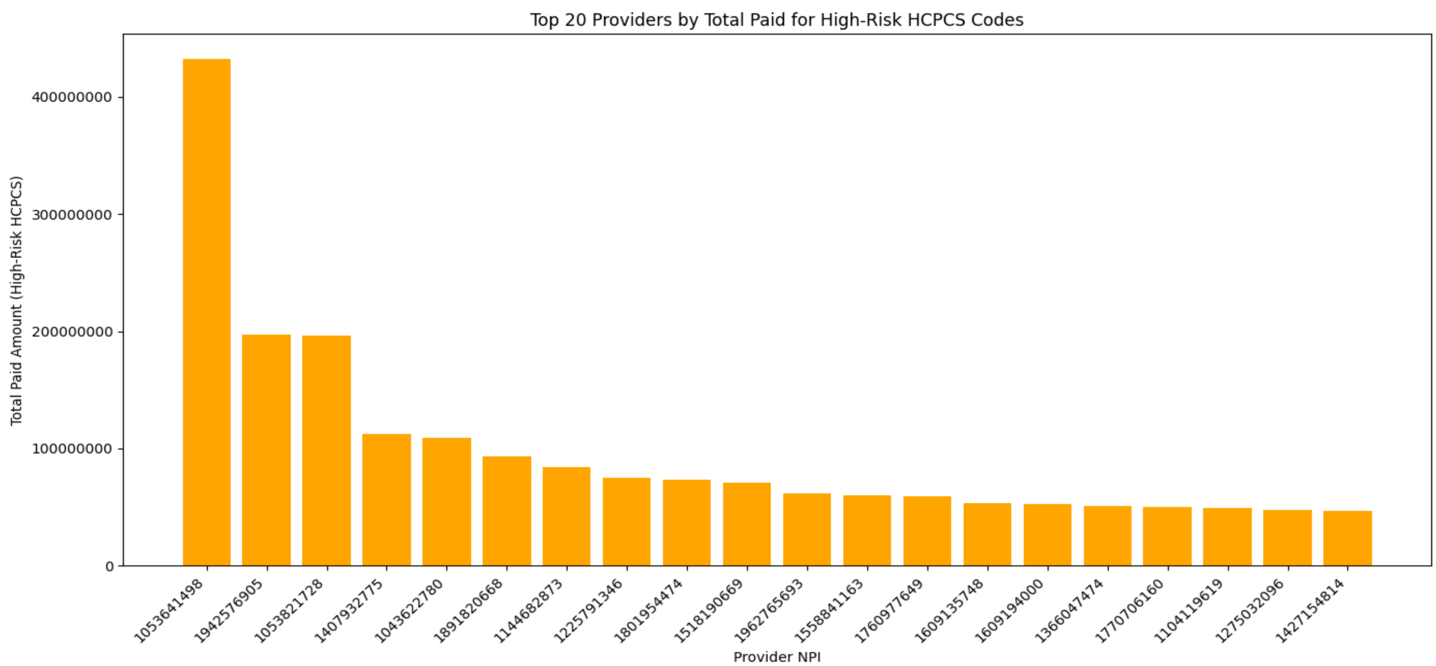
There appears to be a positive correlation between total paid and total claims among the top providers, suggesting that higher payments generally correspond to a larger number of claims for T1019. Similarly, a positive correlation is observed between total paid and total unique beneficiaries for T1019, indicating that providers with higher payments tend to serve more unique beneficiaries. A positive relationship also exists between total claims and total unique beneficiaries for 1019, implying that a higher number of claims is associated with a greater number of unique beneficiaries.

The significant financial volume and concentration around T1019, particularly with the top provider, suggests that this code represents a major service in Medicaid spending.

I narrowed down the analysis to investigate some specific behavioral health service HCPCS code like: 97151, 97152, 97153, 97154, 97155, 97156, 97157, 97158, 0362T.

The HCPCS code had a total claim of 3,9092,187 but with total paid of $7.0217 billion and unique beneficiary of 5,202,139. Further analysis was undertaken to identify provider that were paid for this code. The finding is shown in figure 4 below

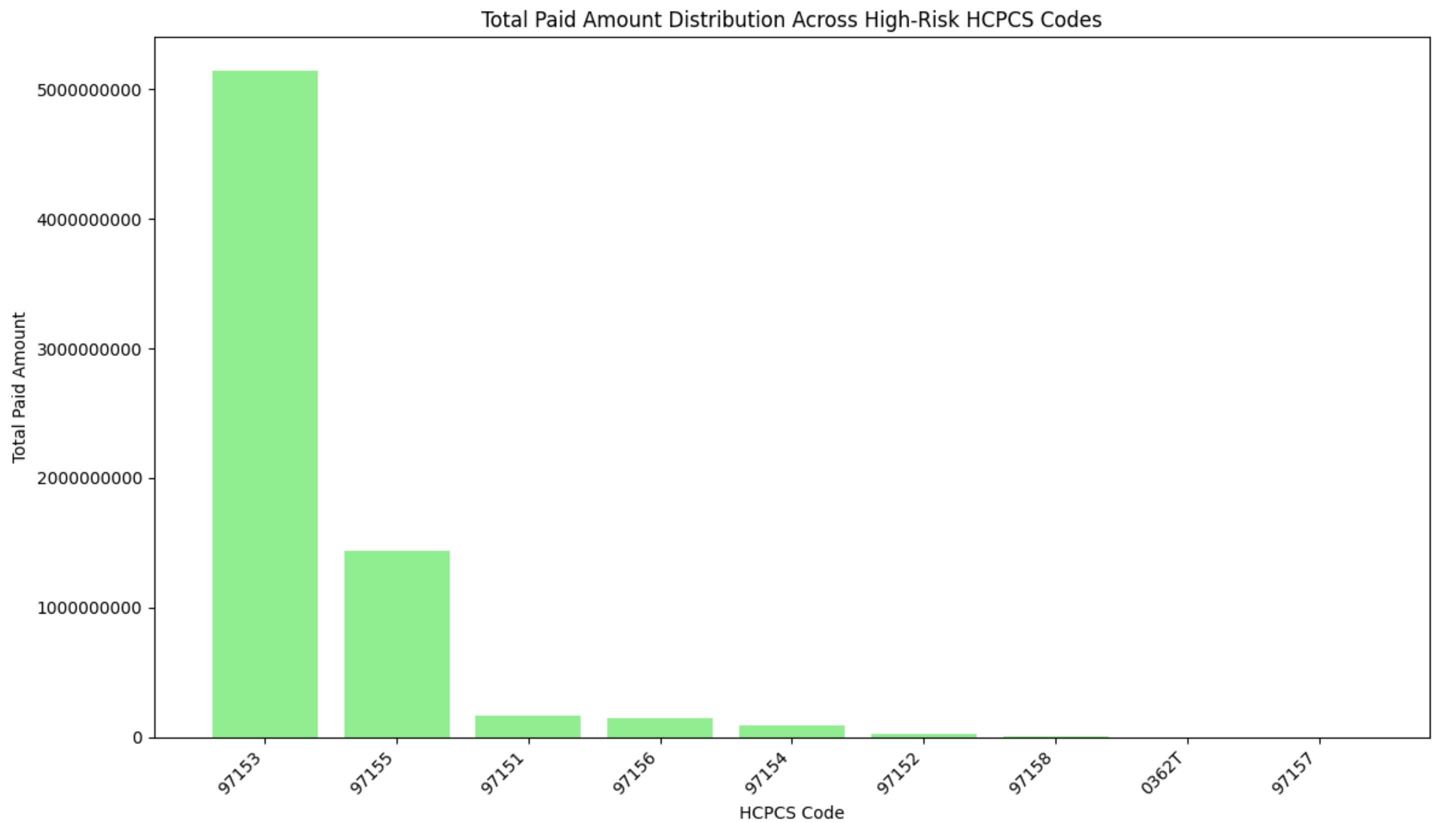
Figure 4 Top 20 providers for specific HCPCS codes



SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date Feb 15, 2026).

Figure 4 above shows a summary of top 20 provider for behavioral health services HCPCS code. Provider with NPI 1053641498 ranked 1 with over $4 billion in payment received. To identify the services which these payments were received for, I carried out further analysis to identify the distribution of total amount paid for the each of the high risk hcpcs code and the result is shown in figure 5 below.

Figure 5. Distribution of Total payment across high-risk HCPCS code

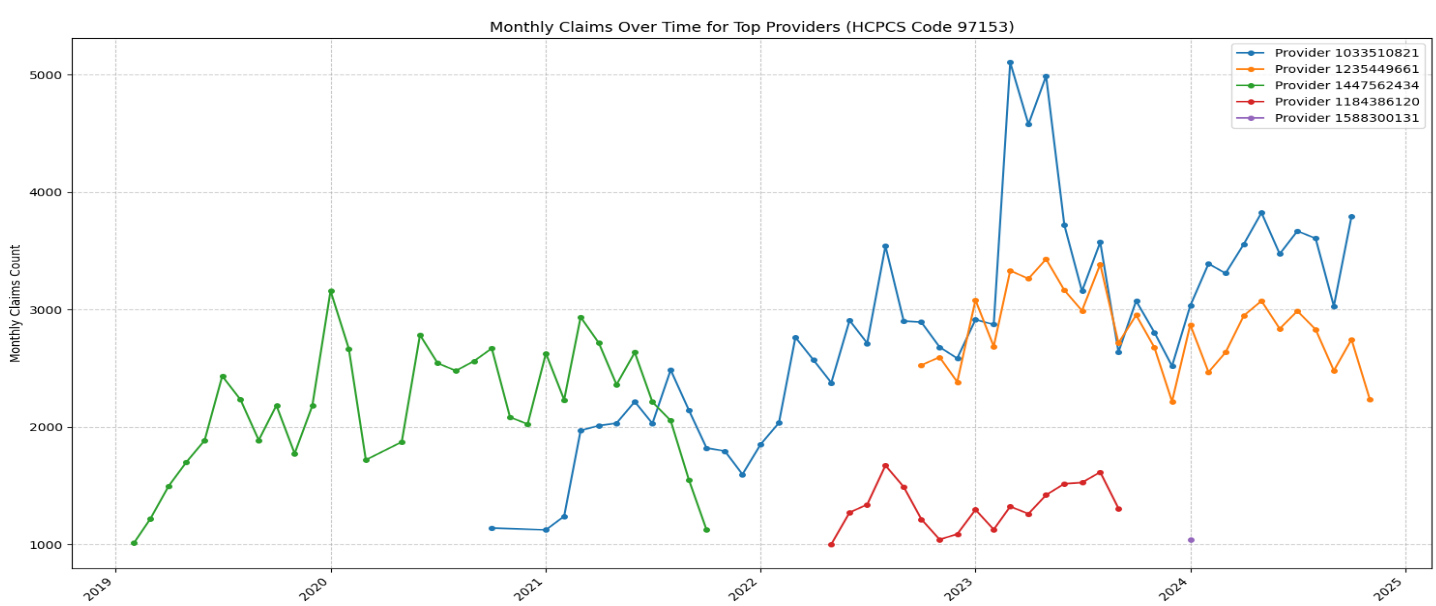


SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date Feb 15, 2026).

Figure 5 above shows a distribution of total paid among the high risk code with HCPCS code 97153 taking a huge chunk of the payment. The total paid amount for the identified high risk HCPCS codes is approximately $7.02 billion, with over 39 million claims and serving over 5.2 million unique beneficiaries. Figure 5 reveals a skewed distribution. HCPCS code 97153 (Adaptive behavior treatment by protocol) is by far the most expensive, accounting for approximately $5.14 billion followed by HCPCS code 97155 (Family adaptive behavior treatment guidance) which is the second most expensive at approximately $1.44 billion.

Further analysis was carried out to analyze the monthly trends for high-risk code 97153 to identify any spikes, seasonality, or sudden changes in spending that might warrant further review. Figure 6 below shows the finding.

Figure 6 Monthly Claims Over Time for Top Providers



SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date Feb 15, 2026).

Figure 6 above reveals a significant billing anomalies in behavioral health and autism related services, particularly under HCPCS code 97153 (adaptive behavior treatment by protocol). Key patterns include extreme monthly claim volumes that suggest unrealistic service delivery capacity, rapid ramp ups followed by sharp declines that is indicative of potential bust out, and volatility among a small set of high volume providers. These findings align with DOGE_HHS X tweet on the 13th of February:

"*For example, using this dataset, it would have been possible to easily detect the large-scale autism diagnosis fraud.*"

Specific behaviors observed for each provider NPI in the chart are Provider 1033510821 (blue line) exhibits the most dramatic pattern, starting with low volumes of 1,000 to1,500 claims per month in 2021, followed by a sharp ramp up in late 2022 and early 2023 to extreme peaks of 5,000 to 5,300 claims per month in mid 2023. It then experiences a steep decline to 2,500 to 3,000 claims per month by late 2023/2024, remaining elevated compared to baseline. This unsustainable growth and drop is a classic red flag for opportunistic or fraudulent billing escalation.

Provider 1235449661 (orange line) displays relatively steady but elevated activity starting around 2022 to 2023, fluctuating moderately between 2,000 to 3,500 claims per month with occasional dips and rises. Unlike others, it lacks extreme spikes but maintains consistently high volumes, potentially indicating a large scale operation warranting review for proportionality to staffing and patient needs.

Provider 1447562434 (green line) shows early high volatility from 2019 to 2022, peaking at 3,000 to 3,200 claims per month in 2020 with frequent ups and downs. It then gradually declines sharply after 2021, dropping to 1,000 to 1,500 claims per month by 2022 and becoming negligible thereafter. This early peak and fade out could suggest a provider that scaled aggressively before reducing or ceasing activity, possibly due to external factors like audits.

Provider 1184386120 (red line) maintains the lowest consistent volumes among the group between 1,000 to 1,800 claims per month with minor fluctuations and no major spikes or drops. It appears more stable overall.

Provider 1588300131 (purple line) appears only briefly at the end (2024 - 2025) with very low volumes between 1,000 claims in a single visible month, showing minimal activity compared to others.

Further analysis to identify outliers providers in the high risk code was carried out.

Table 1

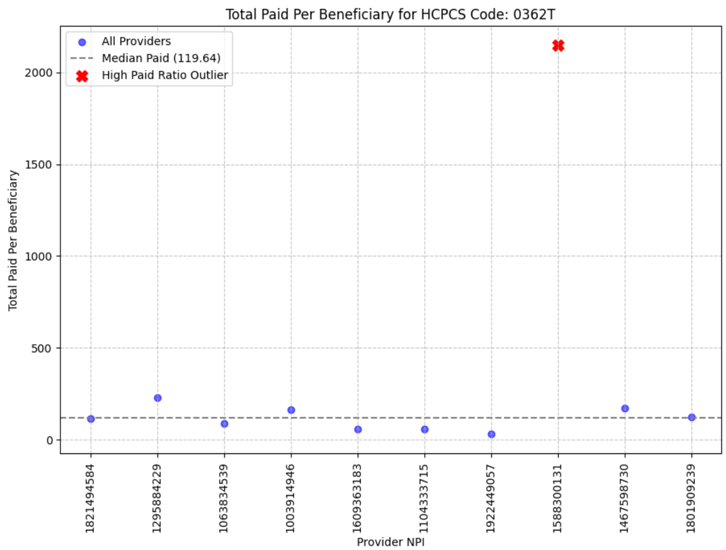
Median per-code metrics for high risk code

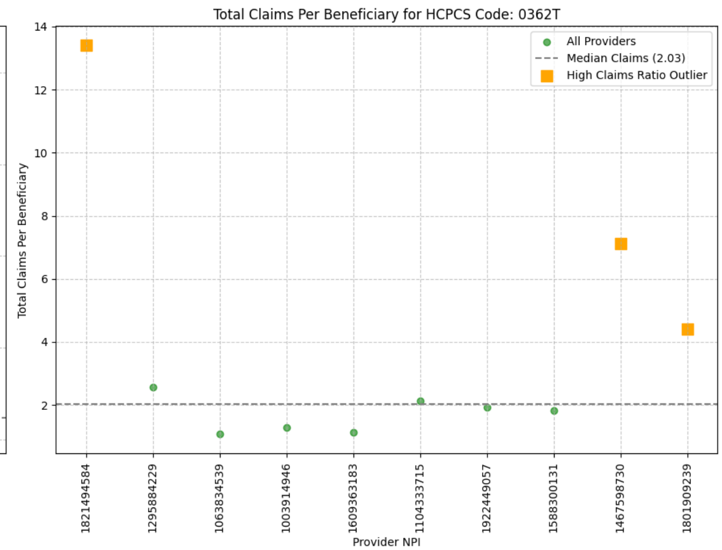
HCPCS Code	Median paid per beneficiary	Median claim per beneficiary
0362T	119.64	2.03
97151	349.62	2.17
97152	310.55	2.04
97153	1888.68	11.54
97154	290.90	6.39
97155	527.05	3.63
97156	180.97	2.04
97157	82.69	1.66
97158	181.06	3.77

SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date March 10, 2026).

Table 1 above shows a summary of of median per code metrics for selected high risk code. The above metrics in addition to paid and claims per beneficiary for each HCPCS code were used to highlight deviations from typical billing patterns. Figure 7 below shows some of the outliers from selected HCPCS high risk code.

Figure 7 Total Paid/Claim Per Beneficiary for Selected HCPCS Codes


Total Paid Per Beneficiary for HCPCS Code: 0362T
All Providers
Median Paid (119.64)
High Paid Ratio Outlier
Total Paid Per Beneficiary
Provider NPI
0
500
1000
1500
2000
1821494584
1295884229
1063834539
1003914946
1609363183
1104333715
1922449057
1588300131
1467598730
1801909239


Total Claims Per Beneficiary for HCPCS Code: 0362T
All Providers
Median Claims (2.03)
High Claims Ratio Outlier
Total Claims Per Beneficiary
Provider NPI
2
4
6
8
10
12
14
1821494584
1295884229
1063834539
1003914946
1609363183
1104333715
1922449057
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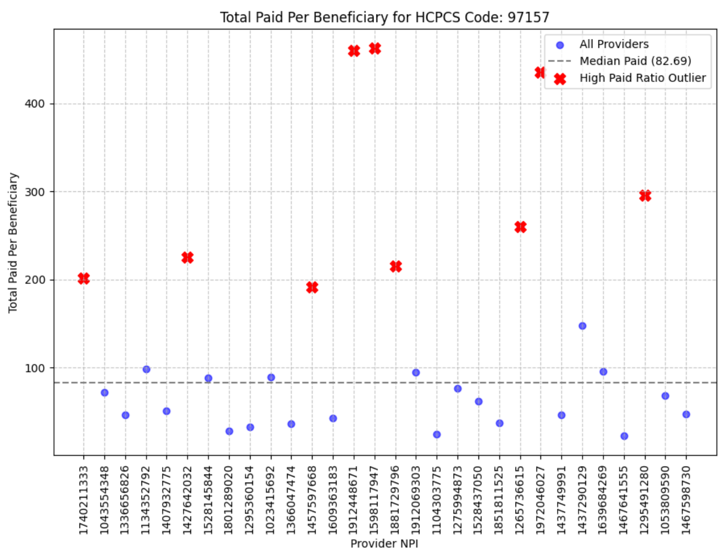
1467598730
1801909239


Total Paid Per Beneficiary for HCPCS Code: 97157
All Providers
Median Paid (82.69)
High Paid Ratio Outlier
Total Paid Per Beneficiary
Provider NPI
100
200
300
400
1740211333
1043554348
1336656826
1134352792
1407932775
1427642032
1528145844
1801289020
1295360154
1023415692
1366047474
1457597668
1609363183
1912448671
1598117947
1881729796
1912069303
1104303775
1275994873
1528437050
1851811525
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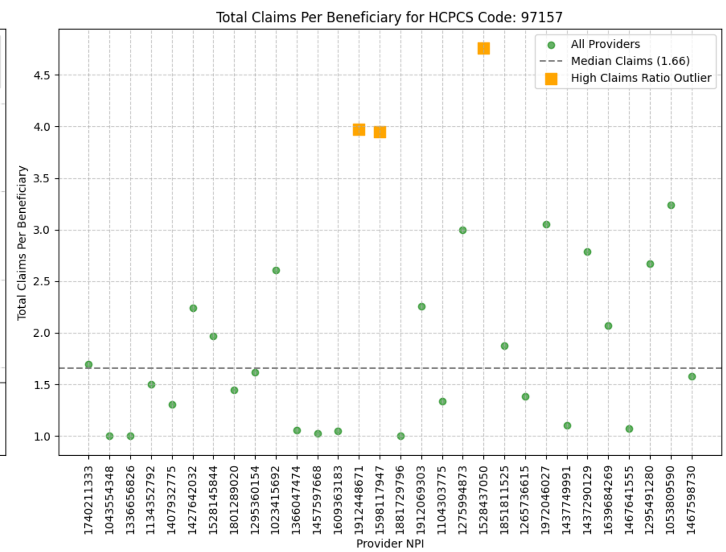
1053809590
1467598730


Total Claims Per Beneficiary for HCPCS Code: 97157
All Providers
Median Claims (1.66)
High Claims Ratio Outlier
Total Claims Per Beneficiary
Provider NPI
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
1740211333
1043554348
1336656826
1134352792
1407932775
1427642032
1528145844
1801289020
1295360154
1023415692
1366047474
1457597668
1609363183
1912448671
1598117947
1881729796
1912069303
1104303775
1275994873
1528437050
1851811525
1265736615
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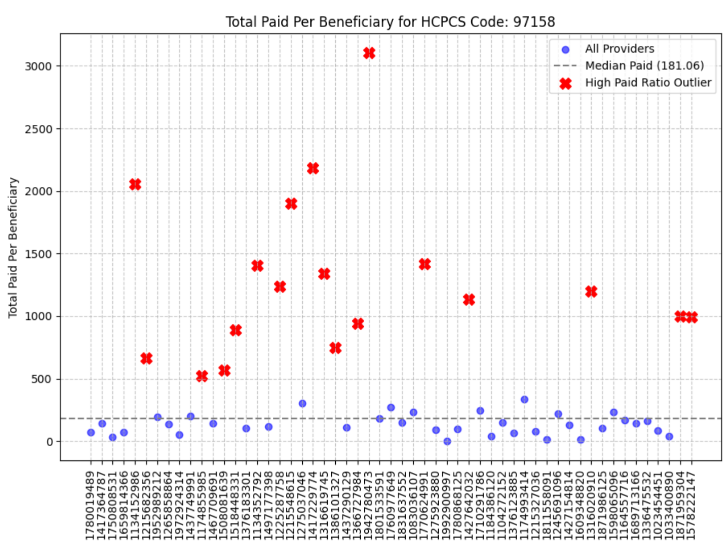
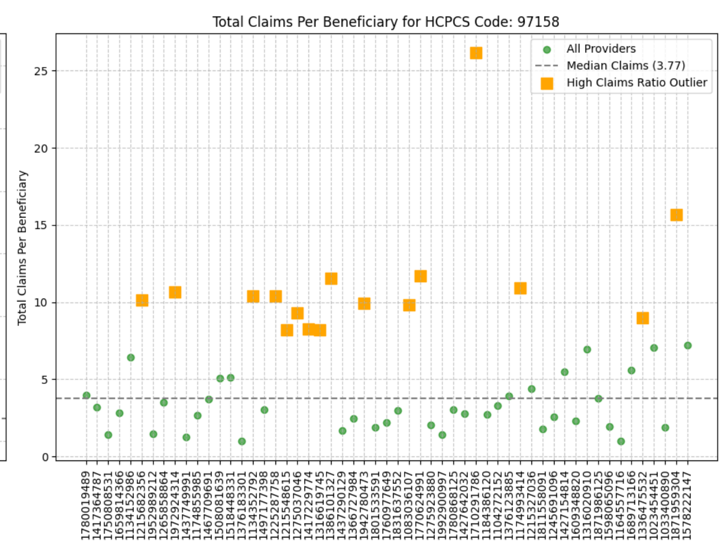
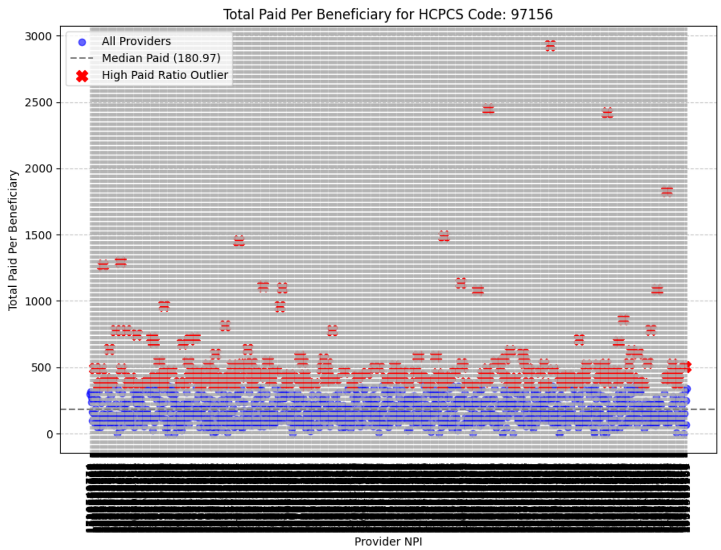
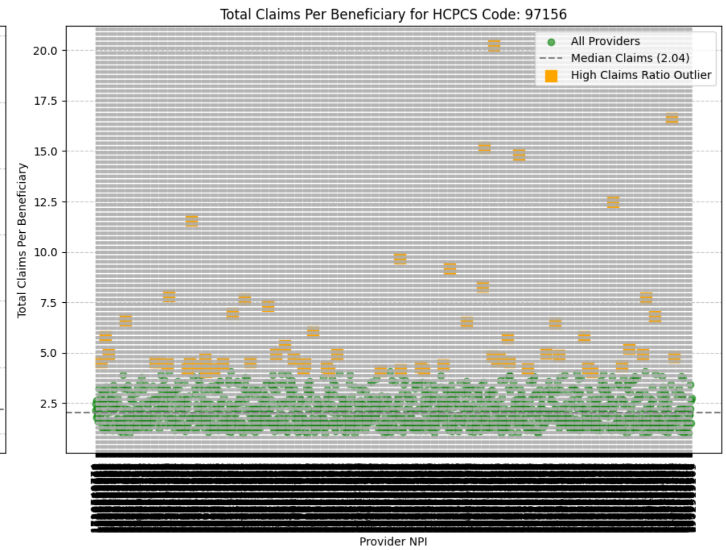
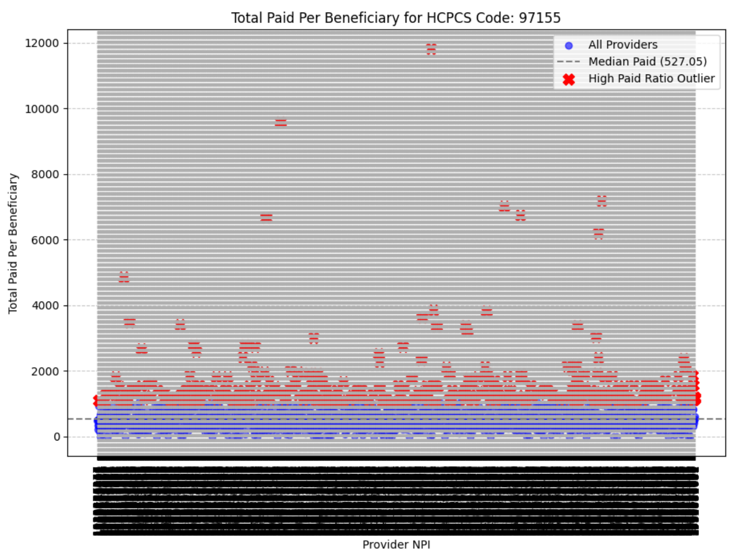
1639684269
1467641555
1295491280
1053809590
1467598730

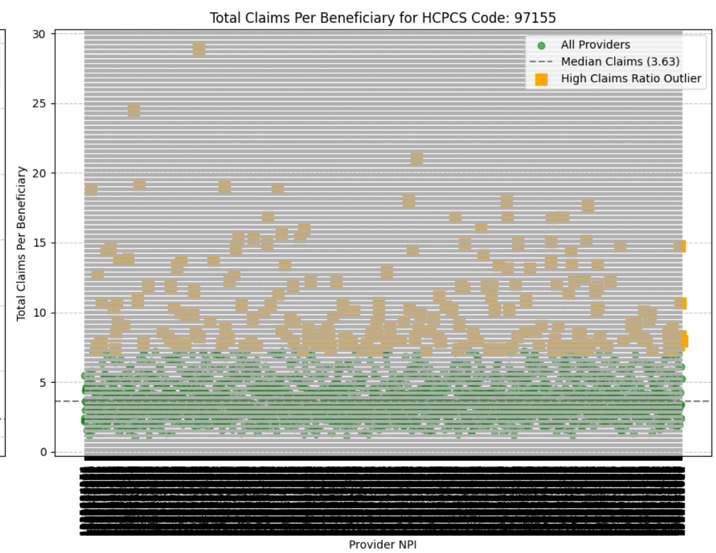

Total Paid Per Beneficiary for HCPCS Code: 97158
All Providers
Median Paid (181.06)
High Paid Ratio Outlier
Total Paid Per Beneficiary
Provider NPI
0
500
1000
1500
2000
2500
3000

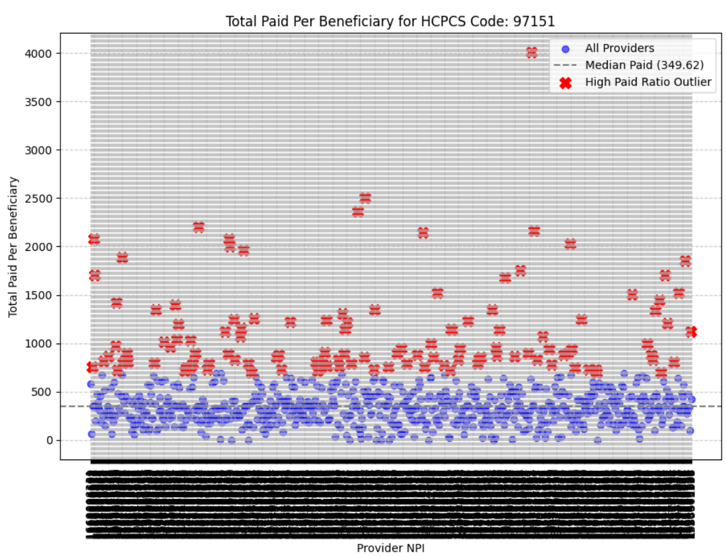

Total Claims Per Beneficiary for HCPCS Code: 97158
All Providers
Median Claims (3.77)
High Claims Ratio Outlier
Total Claims Per Beneficiary
Provider NPI
0
5
10
15
20
25

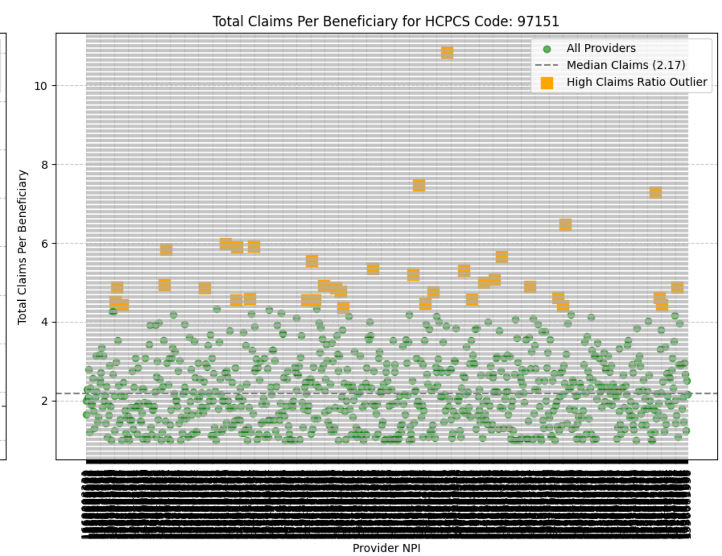

Total Paid Per Beneficiary for HCPCS Code: 97156
All Providers
Median Paid (180.97)
High Paid Ratio Outlier
Total Paid Per Beneficiary
Provider NPI

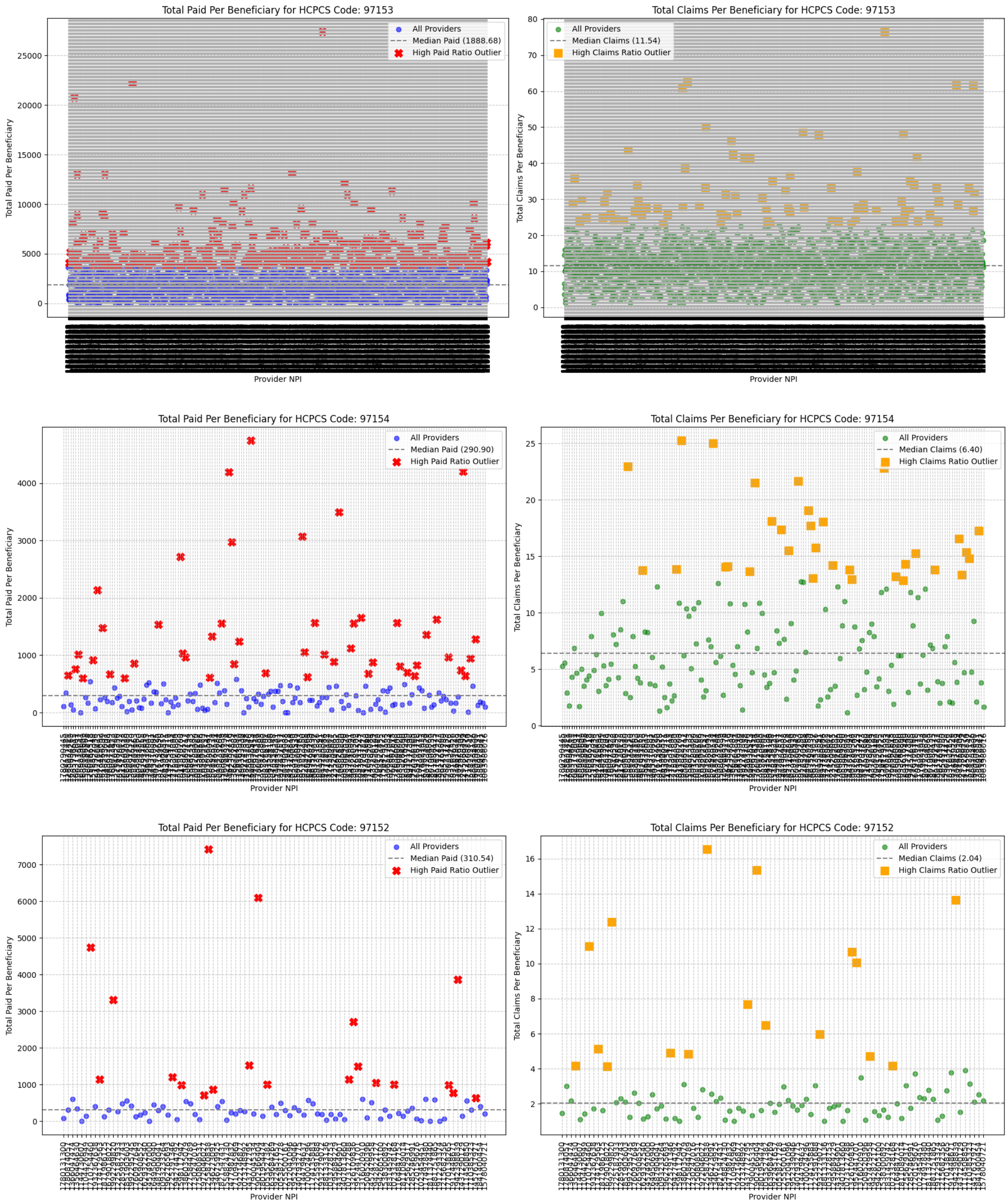

Total Claims Per Beneficiary for HCPCS Code: 97156
All Providers
Median Claims (2.04)
High Claims Ratio Outlier
Total Claims Per Beneficiary
Provider NPI


Total Paid Per Beneficiary for HCPCS Code: 97155
All Providers
Median Paid (527.05)
High Paid Ratio Outlier
Total Paid Per Beneficiary
Provider NPI


Total Claims Per Beneficiary for HCPCS Code: 97155
All Providers
Median Claims (3.63)
High Claims Ratio Outlier
Total Claims Per Beneficiary
Provider NPI


Total Paid Per Beneficiary for HCPCS Code: 97151
All Providers
Median Paid (349.62)
High Paid Ratio Outlier
Total Paid Per Beneficiary
Provider NPI


Total Claims Per Beneficiary for HCPCS Code: 97151
All Providers
Median Claims (2.17)
High Claims Ratio Outlier
Total Claims Per Beneficiary
Provider NPI



SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date March 10, 2026).

The above scatter plots in figure 7 visualize per-beneficiary billing intensity for selected high-risk HCPCS codes but I am going to focus on the first 3 (while other scatter plots provides a clear view of outliers in each code, due to high volume of providers, provider NPI axis is not legible) which are 0362T, 97157, and 97158, comparing individual providers against the median across all providers for that code. They highlight outliers where providers bill far more paid amounts or claims per beneficiary served a strong indicator of potential over servicing, or fraud in behavioral/autism-related services.

In HCPCS 0362T total paid by beneficiary plot most providers cluster very low between 0 and $100 per beneficiary, close to the median line of $119.64. One extreme outlier stands out dramatically at $2,100+ per beneficiary (marked with red *), far exceeding the median and all others. In the total claims per beneficiary plot similar clustering near the median 2.03 claims per beneficiary, with one clear high outlier at 13 to 14 claims per beneficiary (yellow marker).

In HCPCS 97157 total paid per beneficiary, median is 82.69. Several providers are marked as high-ratio outliers (red *), reaching $200 to $450+ per beneficiary and 2.5 to 5x the median or more. In the claims per beneficiary plot, median is 1.66 claims per beneficiary. A few yellow outliers reach between 3.5 to 4.5 claims per beneficiary.

In HCPCS 9715, paid per beneficiary plot median is $181.06. A large number of providers qualify as high-ratio outliers (red *), with values ranging from $500 to over $3,000 per beneficiary and many 3 to 15× the median.

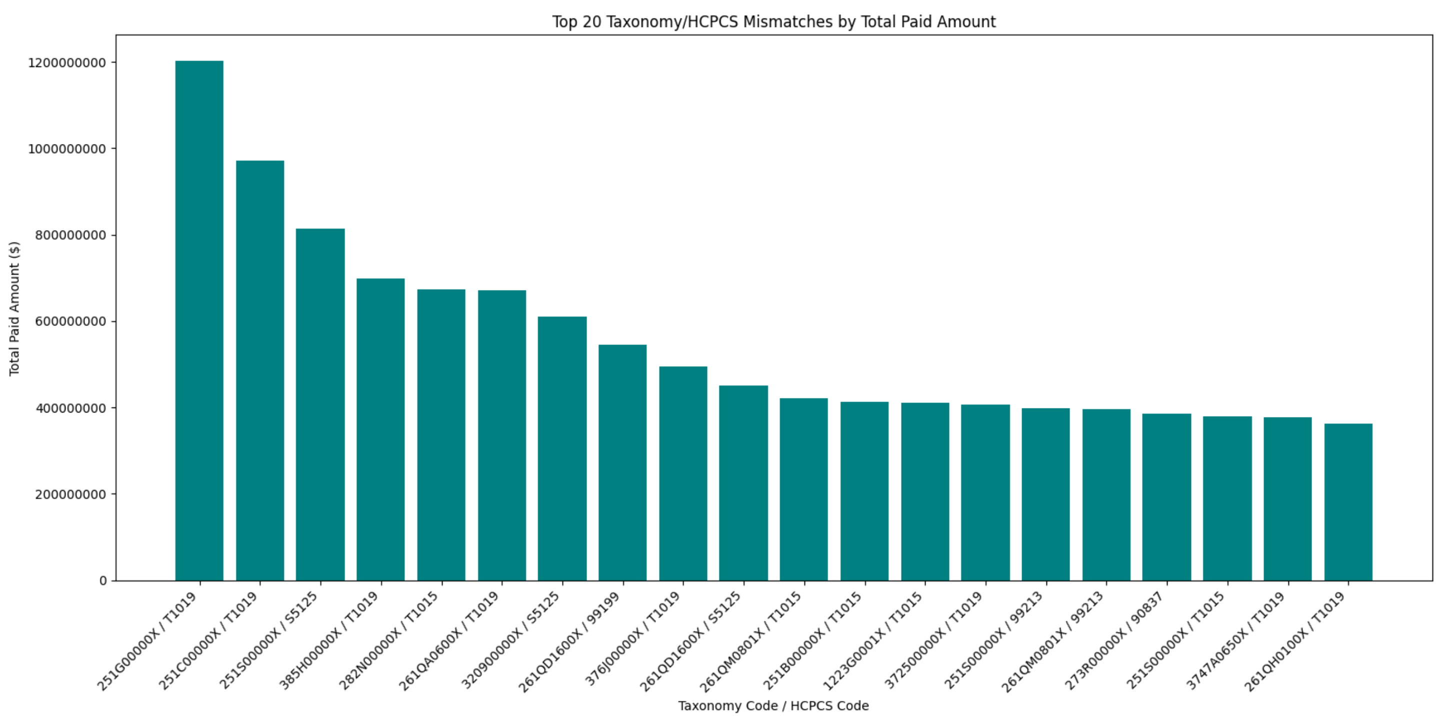
In the claims per beneficiary plot, median is 3.77 claims per beneficiary. Numerous yellow outliers reach 8 to 27+ claims per beneficiary, well above the line.

Finding from the analysis of selected HCPCS code indicate that a lot of providers are billing extraordinarily high amounts and units per patient, suggesting possible inflated service volume or billing without proportional patient benefit.

2. Statistical Discovery of Irregular Billing

More detailed analysis was carried out. This include performing statistical outlier discovery to identify irregular billing patterns where providers categorized as non-specialists who are billing specialty codes that are outside their defined scope or typical practice area. This analysis indicated a total of 291753 irregular billings, and top 20 of those irregular billing by taxonomies are shown in figure 8 below

Figure 8 Top 20 Taxonomy/HCPCS Mismatch



SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date March 10, 2026).

Figure 8 analysis identified Community/Behavioral Health Agencies (taxonomy series 251X) and Clinic/Center-based services (taxonomy series 261X) as the most irregular specialties. These groups frequently billed for high volume codes like T1019 personal care services (Which connects us to figure 2) and S5125 Attendant care services in patterns that were statistically rare (representing less than 1% of total claims for those codes) yet involved massive financial outlays.

The top irregular billing combination, Taxonomy 251G00000X (Community/Behavioral Health) billing for HCPCS T1019, accounted for over $1.2 billion in spending, despite representing only 0.7% of the total claims for that service (see figure 2 above)

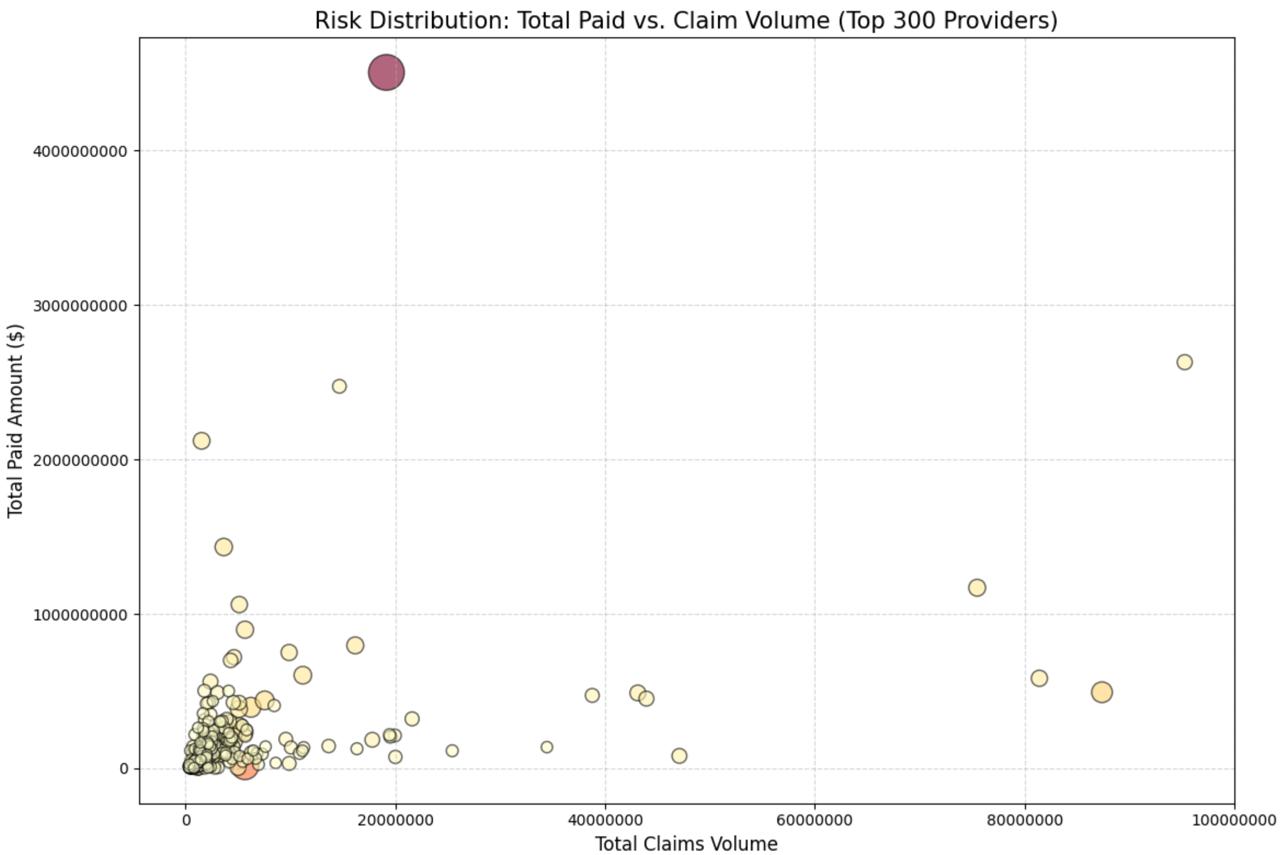
The statistical discovery process identified 291,753 unique instances where a taxonomy's billing for a specific HCPCS code was rare (less than 1% proportion) but significant. Within the top twenty identified mismatches, even the lowest-ranking entry represented over $360 million in Medicaid spending.

Three specific HCPCS codes appeared most frequently in high cost irregular pairings are T1019 Personal care services; T1015 Clinic services; S5125: Attendant care services.

Further analysis to identify the top 300 highest-risk providers based on a combined score of statistical outliers, specialty mismatches, and temporal spikes was conducted. Finding can be seen in figure 9 below together in **exhibit 1.**

3. Geographic Hotspot Mapping

Figure 9 Risk Distribution



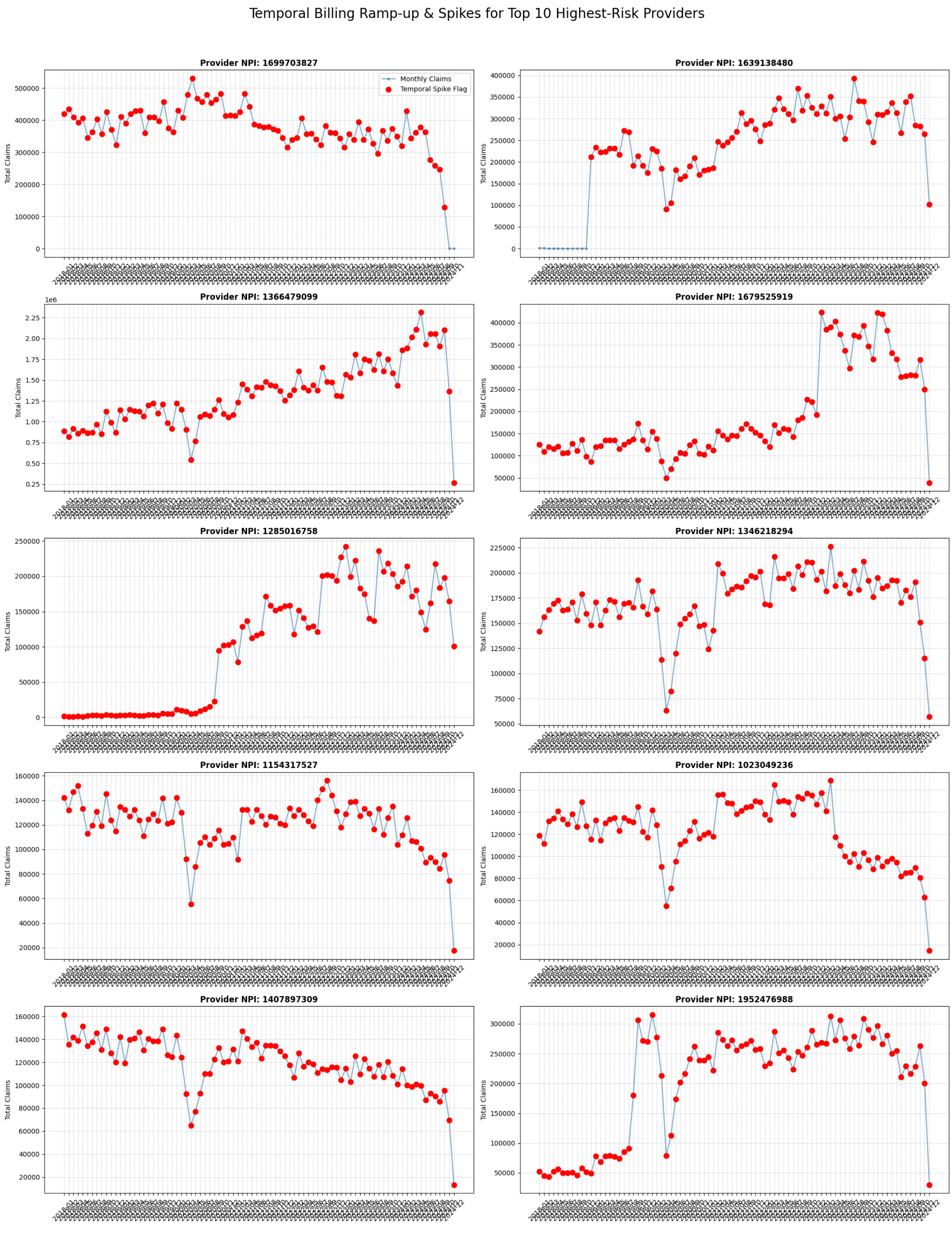
SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date March 10, 2026).

The analysis in figure 9 above conducted a comprehensive risk assessment of Medicaid provider spending by integrating service level data with NPPES provider credentials. I evaluated providers across three key risk dimensions: Statistical Outliers (high payments), Specialty Mismatches (out of scope billing), and Temporal Spikes (unusual volume fluctuations).

Figure 9 and **exhibit 1** show the top ranked provider with NPI: 1699703827 exhibiting over 37,000 risk points, characterized by a massive volume of statistical outliers and temporal spikes, accounting for approximately $4.5 Billion in total paid amount.

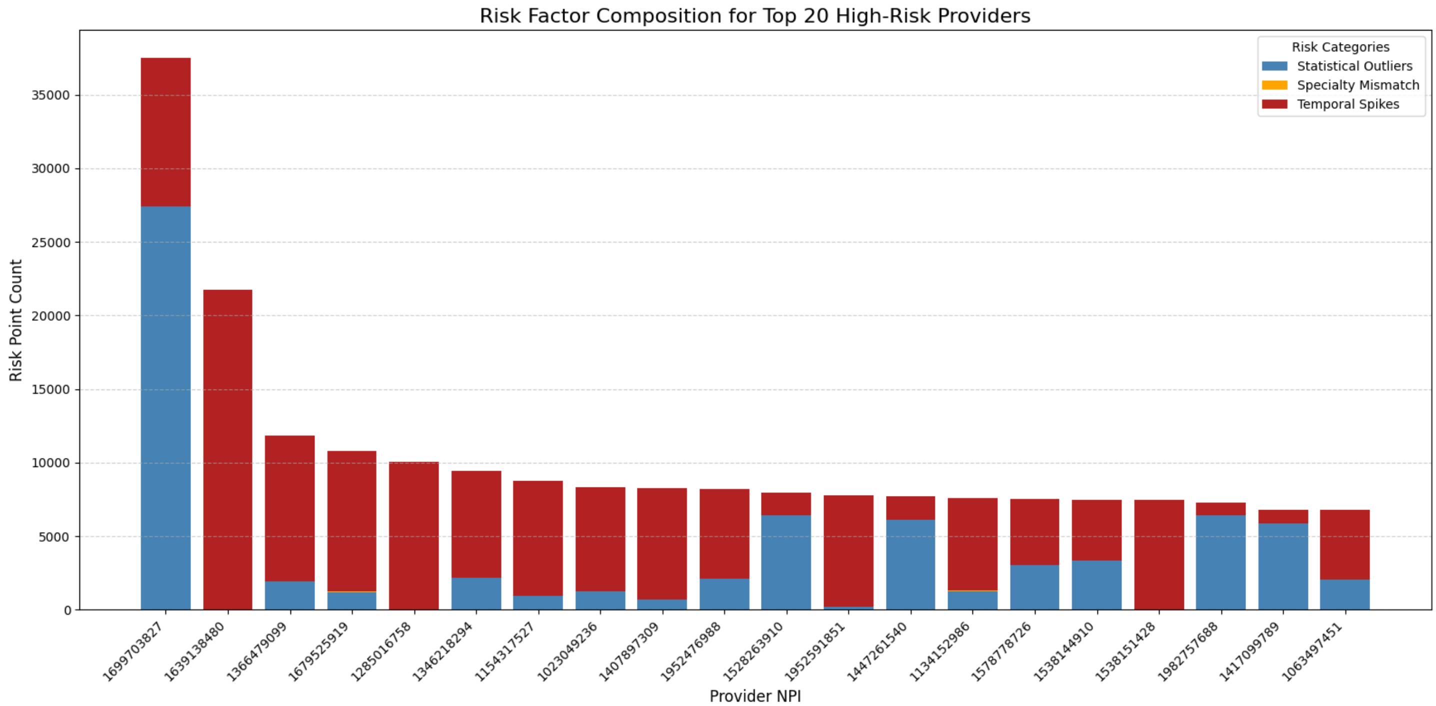
Analysis of the top 300 providers reveals significant risk concentrations in California (Los Angeles, West Hills), Florida (Hollywood), and Ohio (Cleveland) in **exhibit 1**. Figure 10 below shows plots of temporal billing & spikes for top 10 highest risk providers.

Figure 10. Temporal Billing Ramp up and Spikes



SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date March 10, 2026).

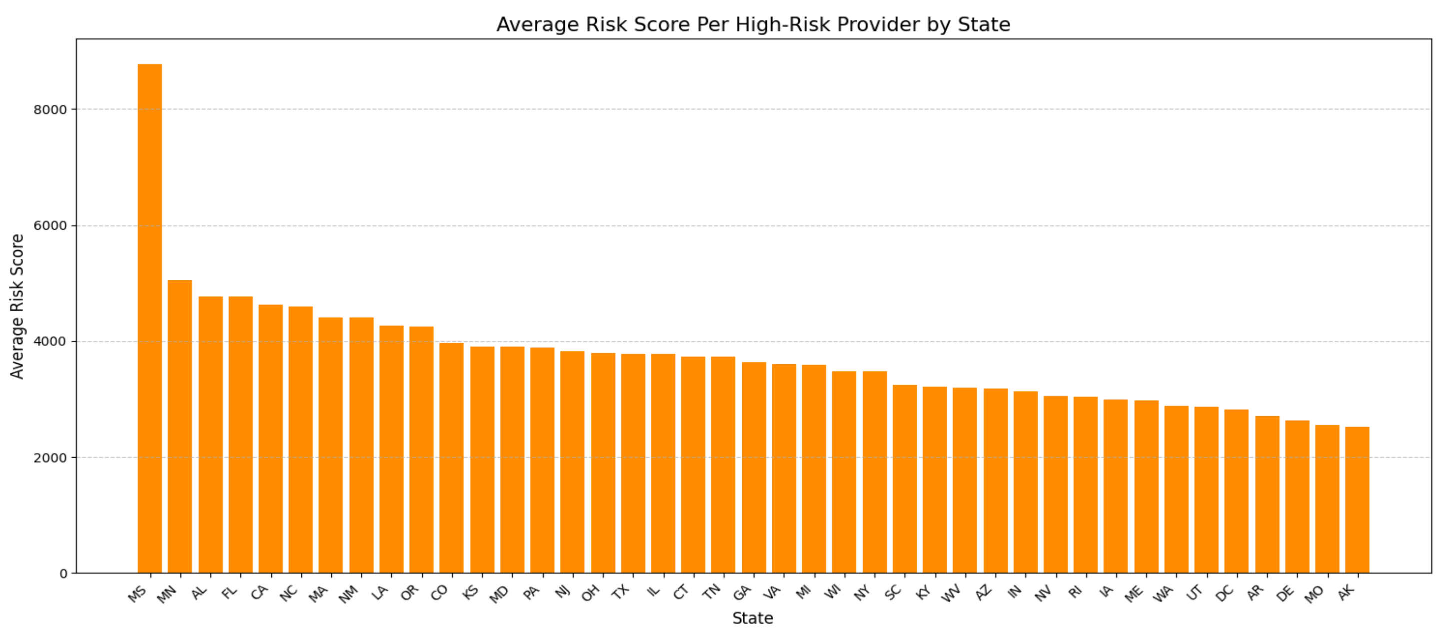
Figure 11



SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date March 10, 2026).

Figure 11 above shows a stacked bar chart that visualize the composition of risk factors for the top 20 high-risk providers, using the pre-calculated flag counts from the ranked evidence table **exhibit 1.** The above figure shows that statistical outliers and temporal strike are the major driver of NPI 1699703827 risk points while temporal spikes are the major driver of risk points of NPI 1639138480. The above figure give a clear direction on area of focus if further analysis and audit are required for each NPI provider.

Figure 12 Average Risk Score Per High Risk Provider by State



SOURCE: Authors' analysis of Medicaid provider spending (2018–2024) Dataset. (Run date March 10, 2026).

Figure 12 above shows the average risk score per high-risk provider across U.S. states, based on metrics like statistical outliers, temporal spikes, specialty mismatches. Mississippi (MS) dominates with the highest average risk score of 8,800, far exceeding all others, more than double the next state. This indicates that flagged high-risk providers in MS have exceptionally elevated combined anomaly scores which include extreme billing intensity, ramps, or mismatches per provider.

Minnesota (MN) ranks second with 5,300 to 5,500 average, followed by a cluster of states in the 4,000 to 4,800 range (AL, FL, CA, NC, MA, NM, LA, OR, CO, etc.). The distribution dropped gradually, with most remaining states falling between 2,000 and 4,000, and lower-risk ones (e.g., AK, MO, DE, AR, DC, UT) at the bottom in the range of 2,000 to 2,500.

Conclusion

The analysis of the newly released 11GB Medicaid dataset reveals a system where financial risk, anomalous billing behavior, and potential fraud are heavily concentrated among a relatively small subset of providers, services. High-risk HCPCS codes particularly behavioral and autism-related services such as 97153, show extreme payment concentration, unrealistic monthly billing spikes, and per-beneficiary intensities far outside normative ranges. These patterns mirror known fraud cases and validate the HHS DOGE team's assertion that the dataset can surface large-scale irregularities.

The investigation also uncovered 291,753 statistically irregular billing events where provider taxonomies billed for HCPCS codes not within their specialty, that represent less than 1% of their typical service mix yet accounted for disproportionately large financial outlays. Community/Behavioral Health Agencies and Clinic/Center taxonomies were the most frequent sources of these mismatches, especially for high-volume codes like T1019 and S5125.

When integrating statistical outliers, specialty mismatches, and temporal anomalies, the risk-ranking model identified a concentrated cluster of high-risk providers, with the top provider alone accumulating over 37,000 risk points and $4.5 billion in total paid claims. Geographic patterns further revealed elevated risk concentrations in states such as Mississippi, Minnesota, California, Florida, and Ohio. Collectively, these findings demonstrate that Medicaid spending vulnerabilities are not random, they are systematic, detectable, and actionable.